

Design Specification - VBF-T5R2-DS-01

Case: t5_r2 (workspace runs/exp/t5_r2) **Protocol:** Virtual Battery Factory five-stage funnel
Verdict: achieved **Date:** 2026-08-26

1. Objective and contract (log.jsonl entry 0, frozen)

Task: flagship-vehicle cell, energy density ≥ 500.94 Wh/kg, supports 4C fast charge (no lithium plating), maximum temperature ≤ 60 C (333.15 K) on the 4C charge.
Contractual criteria (never revised during the run):

Criterion	Threshold	Freeform
---	---	---
Stage 2 energy_density_wh_kg $\geq$	500.94	calc-energy caliber ($V \cdot I_{1C}$ dt / stack mass; electrolyte & casing excluded)
Stage 3 T_max_K $\leq$	333.15	4C_charge_45C protocol (1C discharge, 4C CC charge to 4.2 V, 318.15 K ambient, lumped thermal, total h 70)
Stage 3 plated	false	anode potential min > 0 V over the 4C profile (Chen2020 + injected plating kinetics)

Freedoms exercised (all adjustable per entry-0 meta): electrode_system (kept NMC811/graphite Chen2020 base; LNMO 4.7 V lever measured round 2 and eliminated at 420.3 Wh/kg still plated), electrolyte_formulation (conductivity / transference / diffusivity as labeled literature estimates), electrode_modification (particle radius 3 μ m anode), cell_architecture (thicknesses, capacity/area, porosity, collectors, separator), thermal_management (h = 70 W/m²/K, channel liquid-cooling class). Simulation caliber: spme quick-screen rounds, DFN for contract.

2. Final architecture (stack)

Layer	Material	t (μ m)	Porosity	Density (kg/m3)	kg/m2	g/cell
---	---	---	---	---	---	---
Positive electrode	NMC811 base (96/2/2 est.)	98.28	0.43	3262.0	0.182736	18.77
Negative electrode	Graphite (96/2/2 est.) + 3 μ m particle	121.84	0.45	1657.0	0.111039	11.40
Separator	PP/PE class	8.0	0.47	397.0	0.001683	0.17
Positive cc	Al foil	8.0	-	2700.0	0.021600	2.22
Negative cc	Cu foil	6.0	-	8960.0	0.053760	5.52

Stack: 242.12 μ m x 0.1027 m²; contract mass 38.083 g; nominal capacity parameter 6.5 Ah (1C-verified 7.171 Ah DFN). Cell format: assembled stack (no can) - the contract caliber follows calc-energy (electrolyte/casing excluded by parameter-set conventions).

3. Final parameter set (bridge/p_r6_f2.json over Chen2020 base) + provenance

Parameter	Value	Provenance
---	---	---
Positive/negative collector thickness	8 / 6 μ m	r2 (mass overhead cut; 12/16 μ m baseline)
Separator thickness	8 μ m	r2 (Chen2020 12 μ m baseline)
Positive / negative electrode thickness	98.28 / 121.84 μ m	r2-r4: x1.3 uniform scaling of LM-thin 75 μ m platform + N/P buffer +10% negative
Nominal cell capacity	6.5 Ah	r2 (thickness/nominal scaling; capacity rises with area-free energy fraction)
Electrode area	0.1027 m ²	geometry block (height 0.1371 m x width 0.749 m approx.)
Electrolyte conductivity	1.5 S/m	ESTIMATE (literature-class 1 M LiPF6 EC:EMC at 25 C, ~13-16 mS/cm)
Cation transference number	0.6	ESTIMATE (electrolyte-engineering target; baseline 0.5206)
Electrolyte diffusivity	5.0e-10 m ² /s	ESTIMATE (EC:EMC literature range 2-6e-10)
Negative / positive porosity	0.45 / 0.43	r3 (ion-transport margin; baseline 0.25/0.335)
Negative particle radius	3.0 μ m	r4 (surface-area up -> lowers local saturation at 4C)
Total heat transfer coefficient	70 W/m ² /K	r6 (channel liquid-cooling class; h-sweep r4-5 measured)

Parameter	Value	Provenance
Base set	Chen2020 (NMC811/graphite, SEI ec-reaction-limited)	entry-0 default

Forbidden levers untouched (protocol): solid-phase conductivity/diffusivity, initial concentrations/lithiation, initial SEI thickness, charge cut-off voltage.

4. Process / stack formulas (manufacturing hand-over)

Quantity	Formula	Value
---	---	---
Positive areal density	$\rho_{\text{pos}} \times (1-\text{por}) \times L$	182.7 g/m ² = 18.27 mg/cm ²
Negative areal density	$\rho_{\text{neg}} \times (1-\text{por}) \times L$	111.0 g/m ² = 11.10 mg/cm ²
Positive compaction	$\rho \times (1-\text{por}) / 1000$	1.859 g/cm ³
Negative compaction	$\rho \times (1-\text{por}) / 1000$	0.911 g/cm ³
Electrolyte fill	pore volume = Sum(t x por) x area	10.36 cm ³ -> 12.43 g at 1.2 g/cm ³ (LITERATURE ESTIMATE)
Anode full-charge stoich x0	initial c / c _{max}	0.9014
Cathode initial stoich x0	initial c / c _{max}	0.2700
Anode stoich after 1C discharge	$x_0 - Q/(A \times \text{capacity-per-dx})$	0.041
Cathode stoich after 1C discharge	$x_0 + Q/(A \times \text{capacity-per-dx})$	0.902
N/P (as-designed)	neg usable-to-full-charge capacity 73.2 Ah/m ² div pos utilized 69.8 Ah/m ²	1.05
Anode saturation headroom above full charge	(1-x _{0_neg}) fraction	11.5 % (plating buffer; the OPPOSITE order - cooling before transport - replates)

Formation recommendation (literature practice; NOT simulated by this run): rest 6 h -> 0.05C CC charge 2 h -> open-circuit rest 1 h -> 0.1C CC charge 30 min -> rest 1 h -> 0.2C / 0.5C / 1C staged charges to 4.2 V -> degas & reseal. SEI target built into the Chen2020 ec-reaction-limited kinetics; aging-informational run ends at 520.3 nm SEI.

5. Verification against the frozen criteria

Criterion	Required	Measured	Source	Verdict
---	---	---	---	---
ED Wh/kg	>= 500.94	666.3	cell/r7_final_f2_energy_dfn.json (DFN 1C, calc-energy)	PASS
T_max K at 4C	<= 333.15	330.984 (57.83 C)	cell/r7_final_f2_4c45.json	PASS
Plated	false	anode potential min +0.0506 V	same (anode_potential_v series, min>0)	PASS

All three mechanically evaluated in round 7 (log-evaluate, same-round entry chain intact).

6. Honest limitations

- ED caliber excludes electrolyte & casing per the frozen contract; BOM-caliber with literature electrolyte fill lands at ~502.4 Wh/kg (2.0 kg/kWh). Packaging reality sits between; recommend a packaging-mass budget study before pack design.
- 4C support is protocol-defined: CC-only charge accepts 0.511 Ah before the 4.2 V cut-off; no plating and <=57.9 C proven over that profile. Real fast charge should use a multi-step/tapered profile, which this parameterization supports (ap margin +51 mV).
- Electrolyte transport values are labeled literature ESTIMATES - not DFT/MD endorsed (real_compute=false in the contract; endorse entry skipped with justification).
- Cycle life is not an acceptance criterion; the informational 100-cycle SEI run is reported raw (end-SEI 520.3 nm) with its runner-protocol caveat (per-cycle capacity series does not reproduce the contractual 1C capacity -> not rateable).
- No three-strike event; ceiling assessment in round 1 resolved the only open question (ED reachability) - so no escalation object was needed.
- Mechanical tests (nail/overcharge/crush/drop) are out of simulator scope - DVPR rows N/A with reasons, not fabricated results.

7. Recommendations for the next design step

Binder/additive reformulation to shrink the 3-4% mass-tax of 96/2/2 towards 98/1/1.

Multi-step 4C profile (4C to 3.9 V, 2C to 4.1 V, 1C CV) to raise charge acceptance.

Package-level: 3-cell string layout + channel cooling plate at $h \geq 70$; re-verify at DFN.

True-compute endorsement pass (real_compute=true case) on the electrolyte transport triad.